

Real functions of a real variable



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Real function of a real variable

Introduction: Real function of a real variable

- ▶ In this section we focus on functions whose input and output are **real numbers**
- ▶ We introduce the **graph** of a function, the main tool to visualize how $f(x)$ depends on x
- ▶ We will see that **not all curves in the plane** are graphs of functions

Real function of a real variable

Definition 1: real function of a real variable

A function whose domain D is a subset of $\mathbb{R}$ and whose codomain is $\mathbb{R}$:

$$f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto f(x)$$

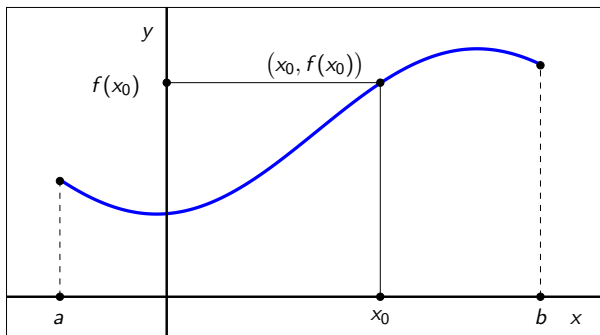
*is called a **real function of a real variable**.*

- ▶ These are functions in which the “input” variable x and the “output” variable $f(x)$ are real numbers.
- ▶ The most common real functions of a real variable have as domain D and as image $f(D)$ an **interval** (possibly the whole of $\mathbb{R}$) or the union of a **finite number of intervals**.

The graph of a function (1/3)

The dependence of the output $f(x)$ on the input x is effectively visualized by drawing the **graph** of f , that is, the set of points of the plane with coordinates (x, y) such that $y = f(x)$ and x in the domain D .

Example of the graph of a real function of a real variable with domain $D = [a, b]$ (a closed and bounded interval):



The graph of a function (2/3)

- ▶ Every line parallel to the y -axis that intersects the x -axis at a point x_0 of the domain D intersects the graph of f at one and only one point.
- ▶ If the line did not intersect the graph, this would mean that no output corresponds to the input x . If the line intersected the graph at more than one point, this would mean that several distinct outputs correspond to the input x , and hence the function would no longer be single-valued.

Therefore, not all curves are graphs of functions.

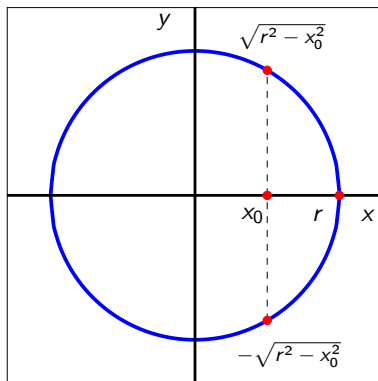
The graph of a function (3/3)

- Note, instead, that nothing prevents a line parallel to the x -axis from intersecting the graph of f at several points or at no point at all.

Example (1/2): curve that does not correspond to the graph of a function

Consider, for example, the curve made of the points of the circle of radius r :

$$x^2 + y^2 = r^2 \iff y = \pm \sqrt{r^2 - x^2}$$



Example (2/2): curve that does not correspond to the graph of a function

- ▶ The vertical line $x = x_0$ intersects the circle at the two points with ordinates $\sqrt{r^2 - x_0^2}$ and $-\sqrt{r^2 - x_0^2}$.

It is not possible to associate a unique output with the input $x_0 \in (-r, r)$, hence this curve does not correspond to the graph of a function!

Real functions of a real variable:

- ▶ $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$: input and output are real numbers
- ▶ The **graph** of f is the set of points $(x, f(x))$ with $x \in D$
- ▶ Every vertical line through a point of the domain intersects the graph at **one and only one point**
- ▶ Not all curves are graphs of functions (e.g. the circle)

Bounded functions

Introduction: Bounded functions

- ▶ In this section we study when the values of a function stay **confined** below, above or between two values
- ▶ We introduce functions that are **bounded above**, **bounded below** and **bounded**, with their graphical interpretation
- ▶ These concepts will be fundamental in the study of limits and of maxima and minima

Definition 2: bounded functions

Given a function $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$, the function is said to be

$$\left\{ \begin{array}{ll} \text{bounded above} & \text{if } \exists M \in \mathbb{R} : f(x) \leq M, \quad \forall x \in D \\ \text{bounded below} & \text{if } \exists M \in \mathbb{R} : f(x) \geq M, \quad \forall x \in D \\ \text{bounded} & \text{if } \exists M \in \mathbb{R}_{\geq 0} : |f(x)| \leq M, \quad \forall x \in D \end{array} \right.$$

Graphical interpretation

► Graphically, we have:

1. a function is bounded above if its graph is contained in the lower half-plane bounded by a line parallel to the x -axis
2. a function is bounded below if its graph is contained in the upper half-plane bounded by a line parallel to the x -axis
3. a function is bounded if its graph is contained in a horizontal strip

Example (1/2): bounded function

Consider the function:

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto \frac{1}{1+x^2} + 1$$

we have

$$\frac{1}{1+x^2} + 1 = \frac{1+x^2-x^2}{1+x^2} + 1 = 2 - \frac{x^2}{1+x^2}$$

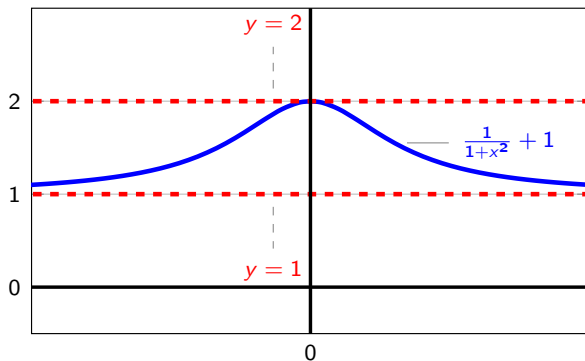
and

$$\frac{x^2}{1+x^2} \geq 0, \quad \forall x \in \mathbb{R}$$

hence

$$1 < \frac{1}{1+x^2} + 1 \leq 2, \quad \forall x \in \mathbb{R}$$

Example (2/2): bounded function

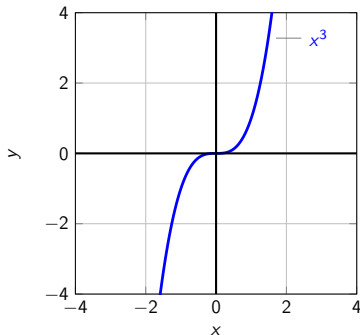


Example: unbounded function

The function

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto x^3 \quad (y = x^3)$$

is bounded neither above nor below.

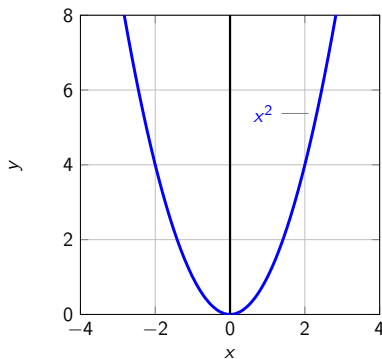


Example: function bounded below

The function

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto x^2 \quad (y = x^2)$$

is bounded below; indeed, $x^2 \geq 0, \forall x \in \mathbb{R}$.



Boundedness and image

- Equivalently, we can say that a function is *bounded above* (*bounded below*, *bounded*) if, respectively, its **image** is a subset of $\mathbb{R}$ that is *bounded above* (*bounded below*, *bounded*).

Summary

Bounded functions:

- ▶ **Bounded above:** $\exists M : f(x) \leq M$ for every $x \in D$
- ▶ **Bounded below:** $\exists M : f(x) \geq M$ for every $x \in D$
- ▶ **Bounded:** $\exists M \geq 0 : |f(x)| \leq M$ for every $x \in D$ (graph in a horizontal strip)
- ▶ **Examples:** $\frac{1}{1+x^2} + 1$ is bounded, x^3 is unbounded, x^2 is bounded below

Symmetric functions



Introduction: Symmetric functions

- ▶ In this section we study functions whose graph has a **symmetry**
- ▶ We introduce **even** functions (symmetry with respect to the y -axis) and **odd** functions (symmetry with respect to the origin)
- ▶ Recognizing a symmetry halves the work when studying a function

Definition 3: even function

*Functions whose graph is symmetric with respect to the y-axis are called **even**.*

- ▶ They are characterized by the relation

$$f(-x) = f(x)$$

which expresses the equality of the ordinates corresponding to the points x and $-x$, which are symmetric with respect to $x = 0$.

Definition 4: Odd function

*Functions whose graph is symmetric with respect to the origin are called **odd**.*

- ▶ They are characterized by the relation

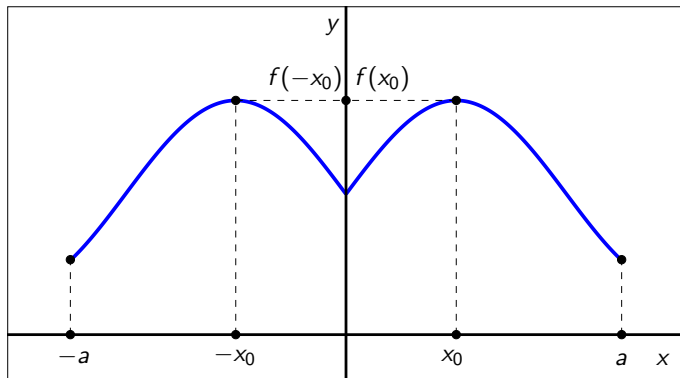
$$f(-x) = -f(x)$$

which expresses the fact that the ordinates corresponding to the points x and $-x$, which are symmetric with respect to $x = 0$, are the opposite of each other.

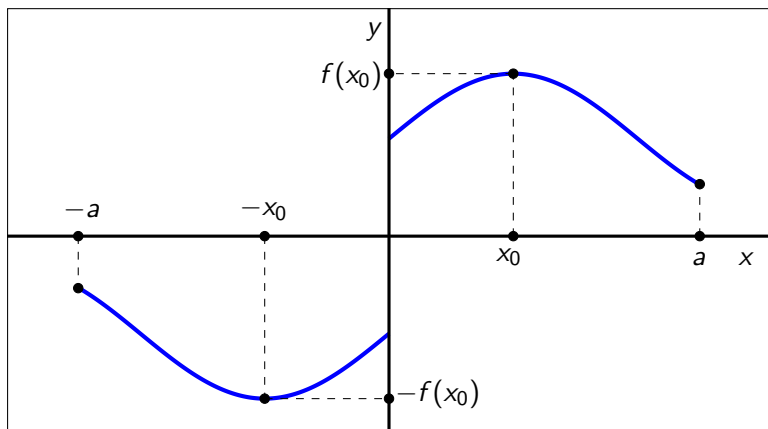
Example: even and odd functions

- ▶ For example, the function $x \mapsto x^2$ is even, while $x \mapsto x^3$ is odd.
- ▶ More generally, powers with integer exponent are even functions if the exponent is even and odd functions if the exponent is odd.

Example of the graph of an even function



Example of the graph of an odd function



No symmetry with respect to the x -axis

A function *cannot* have a graph that is symmetric with respect to the x -axis, since the correspondence would no longer be single-valued.

Summary

Symmetric functions:

- ▶ **Even:** $f(-x) = f(x)$, graph symmetric with respect to the y -axis (e.g. x^2)
- ▶ **Odd:** $f(-x) = -f(x)$, graph symmetric with respect to the origin (e.g. x^3)
- ▶ Powers with integer exponent are even or odd like their exponent
- ▶ No function has a graph symmetric with respect to the x -axis

Monotonic functions



Introduction: Monotonic functions

- ▶ In this section we study how the output of a function changes when the input **increases**
- ▶ We introduce **non-decreasing**, **increasing**, **non-increasing** and **decreasing** functions
- ▶ Monotonicity is one of the fundamental properties in the study of functions and of invertibility

Increasing functions

Definition 5: Increasing function

A function is called **non-decreasing** if for every pair of points x_1, x_2 in the domain of f we have:

$$x_1 > x_2 \implies f(x_1) \geq f(x_2)$$

Definition 6: Strictly increasing function

A function is called **increasing** if for every pair of points x_1, x_2 in the domain of f we have:

$$x_1 > x_2 \implies f(x_1) > f(x_2)$$

Decreasing functions

Definition 7: Decreasing function

A function is called **non-increasing** if for every pair of points x_1, x_2 in the domain of f we have:

$$x_1 > x_2 \implies f(x_1) \leq f(x_2)$$

Definition 8: Strictly decreasing function

A function is called **decreasing** if for every pair of points x_1, x_2 in the domain of f we have:

$$x_1 > x_2 \implies f(x_1) < f(x_2)$$

Interpretation

- ▶ A function f is *non-decreasing* if, as x increases, the corresponding ordinate on the graph of the function does not decrease (hence it either stays the same or increases);
- ▶ A function f is *non-increasing* if, as x increases, the corresponding ordinate on the graph of the function does not increase (hence it either stays the same or decreases).

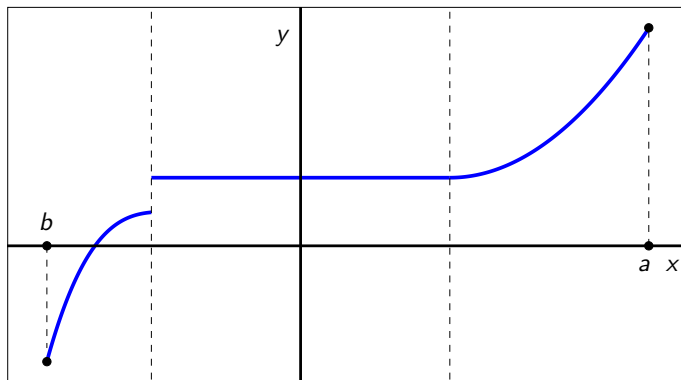
Increasing or decreasing functions are called **monotonic**. Strictly increasing or strictly decreasing functions are called **strictly monotonic**.

Example: monotonic functions

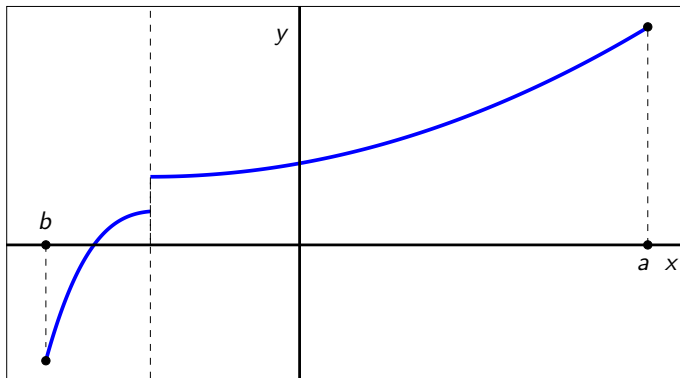
- ▶ For example, the function $x \mapsto x^3$ is strictly monotonic increasing;
- ▶ the constant function $x \mapsto k$ (whose graph is the line with equation $y = k$) is both non-decreasing and non-increasing.

Example of the graph of a non-decreasing function

(horizontal segment)



Example of the graph of an increasing function



- Increasing or decreasing functions (strictly or not) are called **monotonic**.

Summary

Monotonic functions:

- ▶ **Non-decreasing:** $x_1 > x_2 \Rightarrow f(x_1) \geq f(x_2)$; **increasing:**
 $x_1 > x_2 \Rightarrow f(x_1) > f(x_2)$
- ▶ **Non-increasing:** $x_1 > x_2 \Rightarrow f(x_1) \leq f(x_2)$; **decreasing:**
 $x_1 > x_2 \Rightarrow f(x_1) < f(x_2)$
- ▶ Increasing or decreasing functions: **monotonic**; strictly:
strictly monotonic
- ▶ E.g.: x^3 is strictly increasing, a constant is both
non-decreasing and non-increasing

Periodic functions

Introduction: Periodic functions

- ▶ In this section we study functions that **repeat themselves** at regular intervals
- ▶ We introduce the **period** and the **periodicity interval**
- ▶ Periodic functions describe cyclic phenomena (oscillations, waves) and include the trigonometric functions

Definition 9: Periodic function

A (non-constant) function $f : D \rightarrow \mathbb{R}$ is **periodic** with period T , $T > 0$, if T is the smallest positive real number such that

$$f(x + T) = f(x) \quad \text{for every } x \in D$$

- ▶ Every interval of length T contained in D is called a **periodicity interval**.

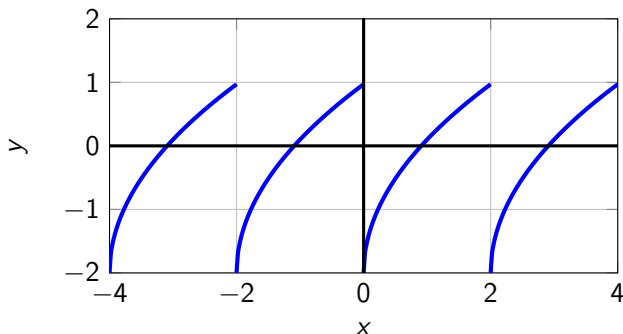
Example: periodic functions

Typical examples of periodic functions are the *trigonometric functions*:

- ▶ $x \mapsto \sin(x)$ ($T = 2\pi$)
- ▶ $x \mapsto \cos(x)$ ($T = 2\pi$)
- ▶ $x \mapsto \tan(x)$ ($T = \pi$)

Example: graph of periodic functions

Graph of a periodic function with period $T = 2$:



Horizontal lines

- ▶ Lines parallel to the x -axis have equation

$$y = k \quad k \in \mathbb{R}$$

Periodic functions:

- ▶ f is **periodic** with period $T > 0$ if T is the smallest positive real number with $f(x + T) = f(x)$ for every $x \in D$
- ▶ Every interval of length T contained in D is a **periodicity interval**
- ▶ Examples: $\sin$ and $\cos$ ($T = 2\pi$), $\tan$ ($T = \pi$)

Final summary

- ▶ A **real function of a real variable** $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$ is visualized through its **graph**
- ▶ **Boundedness**: the graph is contained in a half-plane or in a horizontal strip
- ▶ **Symmetries**: even ($f(-x) = f(x)$) and odd ($f(-x) = -f(x)$) functions
- ▶ **Monotonicity**: (strictly) increasing or decreasing functions
- ▶ **Periodicity**: $f(x + T) = f(x)$, as for the trigonometric functions